Community Deception Techniques: Coordinated Social Engineering in WhatsApp Group Scams

Derek Hansen, Benjamin Schooley, Ryan Rushton, Diego R. Orozco del Cid, Zane Wong, Jonathan Houston
Brigham Young University
Ira A. Fulton College of Engineering
Provo, UT
{dlhansen, ben_schooley, rusht1, orozcod, zw0ng, nonny88}@byu.edu

***Abstract*—Social media messaging platforms have become a common vehicle for cryptocurrency investment scams because they enable fraudsters to create the appearance of legitimate online communities. Although prior research has examined individual social engineering tactics, comparatively little attention has been given to how coordinated groups of fraudulent accounts collectively manufacture trust. This paper presents a mixed-methods case study of two coordinated WhatsApp cryptocurrency scam groups comprising 528 messages and 15–16 fraudulent participant accounts per group. Using qualitative thematic analysis, temporal analysis, and social network analysis, we examine the social roles, interaction patterns, and persuasion strategies used to create the illusion of a thriving investment community. Our findings show that coordinated participants reinforce social proof through complementary roles, dense interaction networks, frequent interactions, and rapid response times while simultaneously leveraging authority, liking, and scarcity. Building on these observations, we introduce the concept of *community deception techniques* to describe manipulation strategies that depend on coordinated interactions among multiple fraudulent accounts rather than the actions of a single scammer. These findings extend existing theories of social engineering and suggest new opportunities for detecting coordinated fraud through community-level behavioral patterns.**

***Index Terms*—community deception, AI deception, social engineering, social proof, astroturfing, whatsapp scams, SNA.**

I. Introduction

Social engineering remains one of the most persistent challenges in cybersecurity because it targets human decision-making rather than technical vulnerabilities [1], [2]. While advances in authentication, encryption, and malware detection have improved defenses against many technical attacks, fraudulent schemes continue to succeed by persuading victims to voluntarily disclose credentials, transfer money, or reveal sensitive information [3]. Most of these attacks begin with phishing or related forms of deception that exploit trust, authority, urgency, and other psychological mechanisms [4], [5]. As communication increasingly occurs through social media and messaging platforms, these environments have become attractive venues for large-scale fraud [6], [7].

Social media fraud encompasses a wide variety of schemes, including romance scams, employment scams, marketplace fraud, impersonation attacks, cryptocurrency investment scams, and "pig butchering" operations [8]–[10]. Although these scams target different populations and employ different narratives, they generally follow a common process based on well-known and long-held social deception mechanisms that leverage technological tools [11]. Attackers first establish a credible persona or attractive opportunity, then cultivate trust or urgency, move the interaction into a communication channel under their control, and ultimately persuade victims to provide money, cryptocurrency, credentials, or other valuable personal information [12]. The effectiveness of these attacks depends less on technical sophistication than on the attacker's ability to create the perception of legitimacy [13].

Messaging applications such as WhatsApp, Telegram, Facebook Messenger, Signal, and Discord have become common facilitators of these schemes because they enable direct communication with large numbers of users while providing features such as encryption, multimedia sharing, and group conversations [14]. Although platform providers continually detect and remove fraudulent accounts and groups, scammers rapidly replace suspended accounts, making enforcement a continual challenge [15]. The low cost of creating or reusing new or fraudulent identities and the ability to reach thousands of potential victims allow these operations to remain both scalable and resilient [15].

A central mechanism underlying many social media scams is social proof, the tendency for individuals to interpret the actions of others as evidence of appropriate behavior, particularly under conditions of uncertainty [5]. Introduced by Cialdini [16] as one of the principal mechanisms of social influence, social proof explains why people are more likely to trust an investment opportunity, product, or organization when they observe numerous others apparently doing the same. Cybercriminals exploit this tendency by fabricating interactions that appear authentic, creating the illusion that many independent individuals have already evaluated and endorsed the opportunity [17], [18]. These campaigns frequently combine social proof with additional persuasion techniques, including authority, liking, reciprocity, commitment and consistency, and scarcity, thereby reinforcing victims' confidence in the fraudulent scheme [16].

Increasingly, these persuasion strategies are implemented collectively rather than individually [19]. Coordinated groups of fraudulent accounts interact with one another to create what appears to be an active, supportive, and trustworthy community. This behavior is closely related to astroturfing, in

which orchestrated activity is disguised as genuine grassroots participation [20], [21], and manufactured consensus, in which attackers intentionally create the illusion that a large community independently agrees on the legitimacy of a product, service, or belief. Rather than rely solely on persuasive messages, attackers create an entire social environment that validates their claims through repeated interactions among seemingly unrelated participants [22]. These persuasion strategies may be further enhanced by synthetic authority, or the perception of credibility created by algorithmic systems, artificial identities, and digitally constructed personas [23].

These coordinated campaigns also represent a form of network manipulation. Instead of treating fraudulent accounts as isolated actors, scammers organize networks of interacting personas that collectively reinforce trust, answer questions, congratulate each other on apparent successes, and model desired behaviors for potential victims [24]–[26]. Previous cybersecurity research has extensively examined phishing, online fraud, social engineering, bot detection, and coordinated inauthentic behavior. However, little work has examined how coordinated accounts collectively construct credibility through their network structure, conversational dynamics, and complementary social roles within private messaging communities.

This paper examines these community-level deception strategies through a case study of a coordinated cryptocurrency investment scam conducted across multiple WhatsApp groups. Cryptocurrency investment scams have become the largest source of reported financial losses in the United States [27], encompassing numerous variants, including fake investment platforms, advance-fee investment schemes, recovery scams, liquidity-mining fraud, and pig butchering scam, in which victims are gradually persuaded to invest larger amounts of cryptocurrency over time [13], [24], [25]. Unlike traditional one-on-one scams, the operation investigated in this study relies on a coordinated community of fraudulent participants who collectively simulate a thriving investment group. Using qualitative analysis, temporal analysis, and social network analysis, we identify the social roles, interaction patterns, and community deception techniques used to manufacture legitimacy and influence potential victims. In doing so, we introduce the concept of community deception techniques to describe manipulation strategies that require coordinated participation among multiple fraudulent actors rather than the actions of a single scammer.

II. Methods

This mixed methods case study evaluates the GateMix cryptocurrency mixing scam operation. A single-case study design is appropriate because it enables an in-depth investigation of a contemporary phenomenon within its real-world context while preserving the complexity of interactions and the setting [28].

A. Data Collection

An author was added to the GateMix Group 742 WhatsApp group chat by an unknown international number on 5/4/2026 and intentionally remained in the group. After a week of messages sent by a variety of accounts, WhatsApp suspended the group. The author was added to a different GateMix 8808 Group on 5/19/2026, which continued for half a week before also being suspended. All 528 messages were downloaded, although only GateMix Group 8808 images were captured since the images could not be downloaded after suspension. Data include chat message content, uploaded images, timestamps, author username (e.g., GateMix), public profile images associated with usernames, and/or phone number (redacted for privacy reasons). The authors also collected information and screenshots from the GateMix mobile app and website.

B. Data Analysis

Social Network Analysis (SNA) and visualization were performed on discussion messages using NodeXL [29] using the WhatsApp data importer. Network nodes represent unique WhatsApp accounts. The directed network edges are based on sequential chat posts, which means sometimes posts were direct replies to the prior post, while other times they were unrelated. For example, if GateMix posts after Miguel, then a line will connect the two with an arrow pointing at Miguel. Edges are also weighted based on the number of times an account replies to the same other account. Messages that are the first post of the day do not count as sequential, so no edge connects them to the prior day's last message. Degree centrality metrics were calculated, including in-degree and out-degree, network density (i.e., percentage of possible edges that are realized), and percent reciprocated edges (i.e., percent of edges that have arrows pointing both directions indicating that both people replied to the other at least once).

A temporal analysis was performed to tabulate and visualize the frequency of posts over time. The median, average, and standard deviation of time between posts made on the same day were calculated.

A qualitative analysis was performed to analyze messages and user roles. Accounts were grouped into four mutually exclusive categories based on an analysis of messages: admin, newcomer, expert, and peripheral. Two authors independently classified accounts in each group. Ratings were identical except for one account, which was listed as peripheral by one rater and expert by another, which was classified as peripheral after discussion. A thematic analysis [30] was also performed to inductively identify the community deception techniques used in the chat group by two authors. They iteratively developed and refined the specific community deception techniques described in Table III by identifying initial techniques with canonical examples, iteratively refining techniques as additional examples were reviewed, combining categories, separating others, and naming the final categories.

C. Ethical Considerations

The WhatsApp groups appeared to be engaged in a coordinated investment scam, although the identities and roles of individual participants could not be independently verified. Messages may have been created by humans or AI agents. Accordingly, all usernames, profile photos, phone numbers,

and other identifying information were modified or obscured. Usernames were replaced with aliases matching the apparent gender, racial/ethnic background, and naming format (e.g., full names, initials) of the originals. Screenshots were edited to remove identifying information except where necessary to illustrate scam features (e.g., the username GateMix is retained). Because WhatsApp has removed the groups, there is no risk of linking the reproduced content to specific accounts.

III. Findings

A. GateMix Crypto Scam Overview

The goal of the GateMix scam is to convince people (i.e., targets) to "invest" cryptocurrency into the scammer's crypto wallet via the GateMix App or website, at which time funds will become unavailable to the victim. Figure 1 shows the overall flow of the scam. It starts by adding victims to a WhatsApp GateMix group where the GateMix admin account tells them about "coin mixing transactions" that "guarantee a 100% return with no risk of loss" since it is "not affected by fluctuations in the cryptocurrency market." A collection of 15-16 likely fake WhatsApp accounts, hereafter called characters, post messages in the group to help answer questions, alleviate concerns, model successful transactions and payments, and encourage each other to visit the GateMix App, website, or contact a Telegram account for more details.

The primary aim of the group chat is to entice the victim to click on a provided link. This link leads to a web application that masquerades as a cryptocurrency investment platform. After registering on the platform, the target is provided with an interface that emulates a legitimate investment application. They are incentivized to carry out the following actions:

1. Deposit Funds: By sending cryptocurrency to GateMix's wallet, their web account appears funded. However, the Crypto assets actually sit in the wallet of the scammer–or may already be moving through a laundering system.

2. Verify with Identification: The target can "verify" their account by uploading a photo of their license, passport, or other government-issued identification. The scammers can then use these documents to commit identity theft.

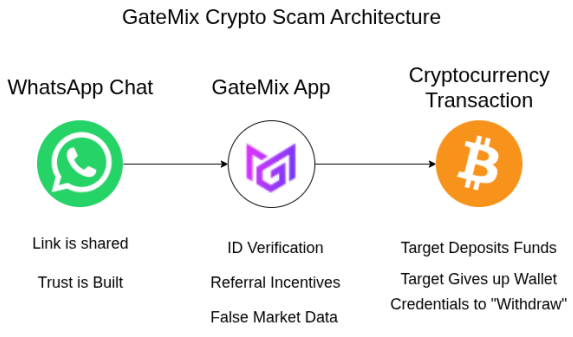


Fig. 1. GateMix Crypto Scam Architecture

3. Recruit Friends: The target is motivated to gain rewards by referring their own friends to the site.

4. Withdraw "Profits": The target is prompted to enter the credentials of their crypto wallet to receive payments. These credentials can be used by the scammer to drain the wallet and attempt to sign into other accounts owned by the target.

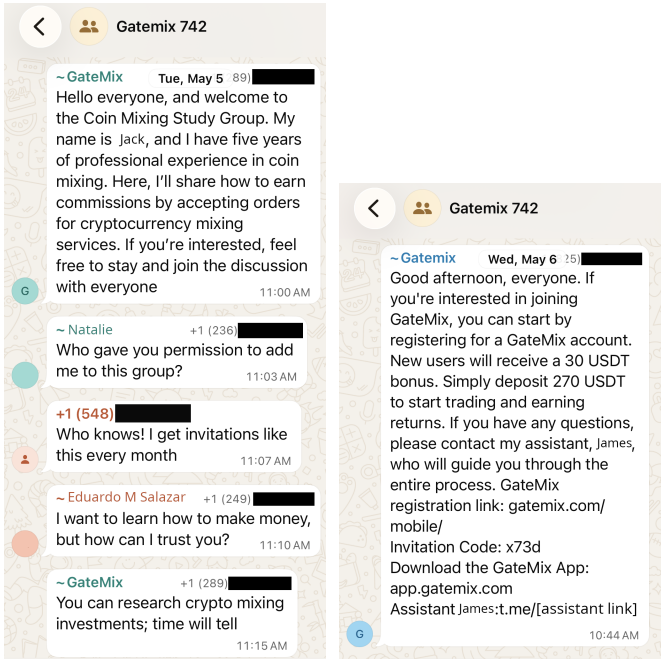
There is high confidence that this is a scam for several reasons. The website only accepts cryptocurrency deposits, not other forms of credit. Both WhatsApp groups analyzed in this paper were suspended shortly after creation and many character profile images were changed to other groups likely associated with scams. The website and mobile app do not follow proper security protocols (e.g., CAPTCHA doesn't require proving you're human; identity verification doesn't use a secure portal). Finally, as described in the remainder of the paper, the messages themselves are quite suspicious.

B. WhatsApp Group Setup and Management

1) An account (which is later deleted) creates a GateMix group (e.g., GateMix Group 8808, GateMix Group 742) with end-to-end encrypted communication turned on. This may suggest a role of someone who creates new WhatsApp groups and invites scammers or Large Language Model (LLM) agents who take on other roles.
2) International accounts from non-U.S. regions (Pakistan, India, Yemen, Indonesia) automatically add U.S.-based targets to the group in large batches (e.g., 25-97 accounts). This suggests a scammer role is to identify WhatsApp target accounts to add to the groups. These international accounts joined using a group link, suggesting coordination outside the WhatsApp group.
3) An international account adds what we believe are fake U.S. accounts that act as characters.
4) (optionally) A U.S. character account removes all international accounts from the group.
5) A U.S. character account changes group settings from allowing only admins to post to allowing anyone to post. It also changes the group description and/or name.
6) (optionally) The official GateMix account removes the account that changed the group settings and accounts of targets who indicate that this is a scam. For example, an account in GateMix Group 8808 posted "Scam" over a dozen times before any welcome messages had been sent. GateMix deleted the account within an hour and a half, showing its moderation role.
7) (optionally) If the group is flagged as Suspended or Deleted then members may be added to another group, which explicitly mentions that it is a "new cryptocurrency mixing and trading group" including "both new and experienced members." This suggests that the same group leadership operates across multiple groups, despite the fact that the phone number assigned to the GateMix account changed.

TABLE I
STATISTICS FOR GATEMIX GROUP CONVERSATIONS.

	GateMix Group 742	GateMix Group 8808
Total posts (including setup)	352	176
Active Days	7 (starting 5/5/2026)	3 (starting 5/20/2026)
Character accounts	15	16
Avg. Character posts per day	48	44
Images Shared	65 (19% of posts)	32 (24% of posts)
Median time between posts	3.8 minutes	3.2 minutes
Avg time between posts	9.1 minutes	10.8 minutes
Std dev. time between posts	15.7 minutes	21.6 minutes



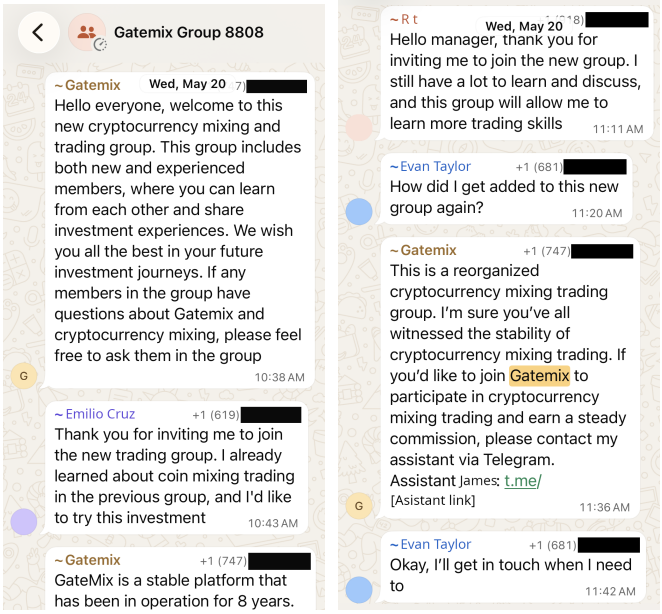
(a) Welcome Message (b) Daily Reminder

Fig. 2. Introductory Messages

C. GateMix Discussion Overview

The discussion between fake accounts (i.e., characters) is similar in each of the two WhatsApp GateMix group chats we examined (Table I). All posts in the chats are from the 15-16 characters, with one exception where a target repeatedly posted "spam" in GateMix Group 8808 before being removed. The groups begin with the admin account GateMix explaining the purpose of the group and sharing information on how to get involved (see Fig. 2a and Fig. 3). An explanatory message by GateMix is repeated at the start of each day (see Fig. 2b).

Conversations continue as different characters post questions and answers, provide guidance and instruction, discuss profits made, and inspire others to invest. Characters often reply to posts in ways intended to build trust, even among the skeptical (see Fig. 2a and Fig. 3) and reinforce the benefits of participation (see Fig. 3b). Characters post screenshots of money they have made (see Fig. 4) or evidence that they have withdrawn money successfully, constituting from 19-24% of posts. They also ask and answer questions and share advice.



(a) Migration Welcome (b) Explanation to reinforce authority and appease critics

Fig. 3. Messages in new chat after original chat was suspended

D. Social Roles

Four primary social roles were identified in the GateMix communities: admin, experts, newcomers, and peripheral members. Table II indicates the charcters from GateMix Group 8808, their role, and a short profile. The characters and their interactions are designed to persuade targets to trust and invest in GateMix and resolve perceived concerns or questions. For example, newcomers often ask questions that experts reply to. They also model joining the community and becoming an active participant. Meanwhile, experts frequently share earning, discuss how reliable GateMix is, and how there is "absolutely no risk" with this type of investment. They set the standard that the rest of the group members attempt to model, including potential victims. The administrator (i.e., GateMix account) introduces the GateMix investment opportunities, earning potential, answers questions, and frequently sends messages regarding how to sign up and get started. He congratulates the other members of the group on their earnings and encourages others to get started. He reminds us to reach out to his assistant, James. Although James is not a member of the group, his Telegram contact information is shared frequently and experienced investors refer to his help. Peripheram members are those who only post 1-2 messages, such as a question or response, but don't get involved heavily.

All characters in both groups have U.S. phone numbers. Profile pictures in GateMix Group 8808 indicate a variety of ethnicities and ages (though all were adults), with only 25% (4/16) women. The diversity is likely intentional so targets can relate to similar people. All accounts include a

username (e.g., Megan) except two newcomers who only list a phone number. The only member without an image is also a newcomer, perhaps helping targets relate more to the persona since no targets had public profile pictures. There were more missing profile images in GateMix Group 742 because data was collected after a longer delay period and images were removed or modified after group suspension.

TABLE II
CHARACTER ANALYSIS FOR GATEMIX GROUP 8808

Name	Role	Profile
GateMix	Admin	Moderates discussion and directs to scam site
Emilio cruz	Newcomer	Experienced investor, new to GateMix
Gary collins	Expert	Peer mentor who guides newcomers
R t	Newcomer	Excited newbie, impressed by everyone's earnings
Evan taylor	Peripheral	Asks why he was added to group
Miguel	Expert	Models investing and posts about earnings
Eric barnes	Expert	Advisor and model investor
+1 (661)	Expert	Answers questions, displays evidence of success
Elena	Newcomer	Impressed by how easy it is to invest
+1 (626)	Newcomer	Ask questions on how crypto trading works
Alina	Expert	Reinforces how everyone working as a team
Bobby	Peripheral	Claims to have processed 3 orders
+1 (810)	Newcomer	Enthusiastic new investor
Megan	Expert	States that the investment is risk-free
Jason miller	Expert	Responds to questions
Marco soto	Peripheral	Responds to a question from Emilio

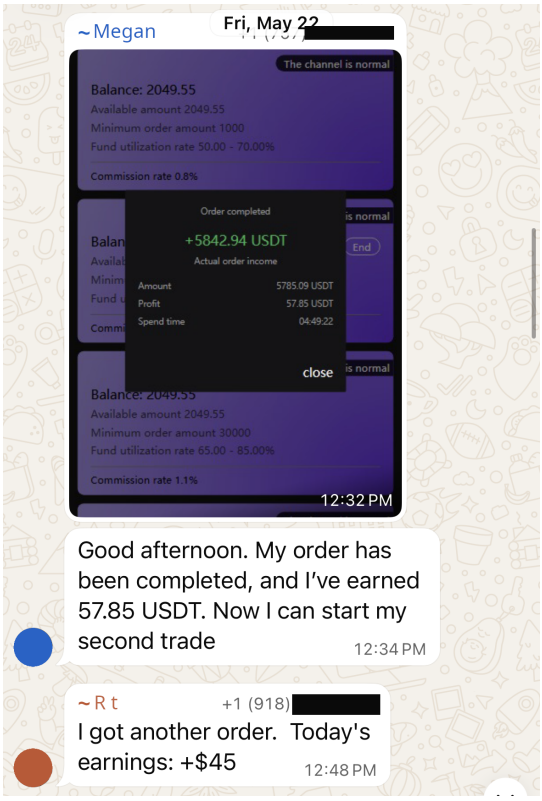


Fig. 4. Megan posts a screenshot of the GateMix website showing her successful profit of $57.85 as evidence that the system works, with R t reinforcing his earnings 14 minutes later.

E. Network Analysis

Both GateMix Group 742 (see Fig. 5) and GateMix Group 8808 (Fig. 7) showed strikingly similar network characteristics. They included a similar number of characters (15 and 16), including within each role (e.g., 5-6 newcomers, 6-7 experts, 1 admin, and 2-3 peripheral accounts). The networks are fairly dense (0.26 density for group 8808 and 0.45 density for group 742), reflecting the fact that many accounts post consecutively after other accounts as seen in Fig. 5 and Fig. 7. This likely contributes to the sense of community, since different, but familiar characters post consecutively, often replying to each other in ways described later. The percent of edges that are reciprocated (i.e. point both directions indicating each user replied to the other at least once) is also high (78% for group 742 vs 44% for Group 8808), indicating relatively frequent back and forth posting. This helps convey a feeling of responsiveness between individuals.

The GateMix account is connected with all characters except one (in each group), showing its central role as administrator. Most experts post frequently, while newcomers vary in their posting frequency. There are examples of newcomers (e.g., Emilio Cruz or Tommy Navarro) with high in-degree and out-degree centrality measures indicating their central role in the network. These newcomers transform into experienced investors over time. For example, Tommy Navarro starts by saying: "I like making money, so I'll stick around to observe and learn; it's a good thing to pick up more money-making skills." After learning from others and posting about his initial successes, he ends by posting: "My orders are all complete. I'll be able to finish all 20 of them by Monday. I need to carefully plan how to allocate my funds for future investments" and then shares screenshots of his success.

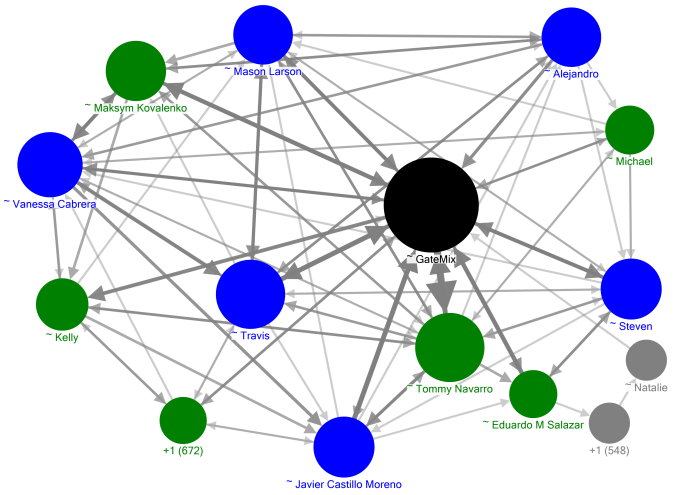


Fig. 5. Network graph of GateMix Group 742 characters. Vertices show public profile images that are sized based on total posts to the group and are colored by role: admin (black), newcomer (green), expert (blue), and peripheral (grey). When an account (e.g., Miguel) posts directly after another account (e.g., GateMix), on a given day, a directed edge is shown pointing at the first poster (e.g., GateMix). Edge width and opacity are based on the number of consecutive posts between the two accounts.

In summary, the network models a thriving community with connections dispersed among most accounts including experts and newcomers, some of whom transform into experienced "investors." This network structure fosters a feeling of an active community of mutual learning and validation.

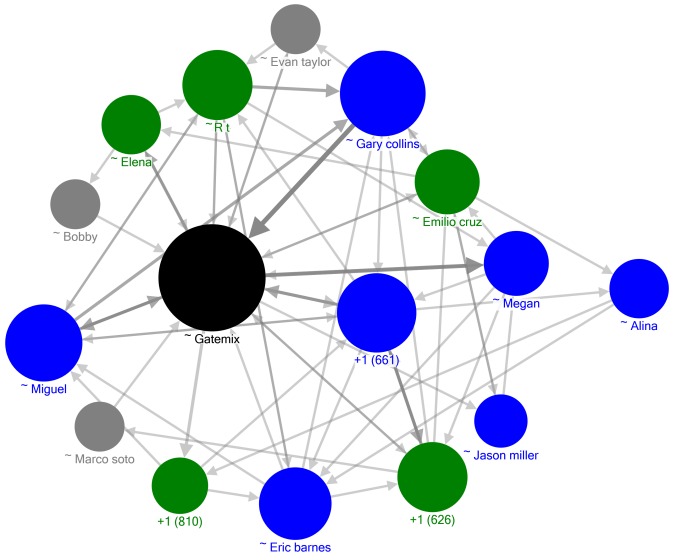


Fig. 7. Network graph of GateMix Group 8808 with same visual properties as Fig. 5

F. Temporal Analysis

As shown in Fig. 6, most characters participate throughout each day. This helps convey a responsive and active community with familiar faces reoccurring at different intervals. Posting occurs frequently, primarily during the hours of 11am-4pm Mountain Time from Tuesday, May 5th, 2026 through Monday, May 11th, 2026 (excluding Sunday, when no messages were posted). This time range covers times when people are awake in most U.S. time zones. The time between posts is quite rapid, with an average of 9.1 minutes (GateMix Group 742) and 10.8 minutes (GateMix Group 8808) between messages. This rapid response time with infrequent lag times conveys a sense of a thriving, responsive community.

G. Textual Analysis

The text itself has several strange properties that indicate that it is likely generated by an LLM or a template. First, there is a very consistent voice for nearly all of the characters which is polished, overly enthusiastic, and strangely promotional. For example, Alina's introductory post reads: "Manager, I am honored to continue learning about coin mixing with everyone on this new team" (no period). Another post by Vanessa Cabrera reads: "My order has been completed, and I'm planning to withdraw some of my profits so I can enjoy a wonderful weekend. Thanks to my investment in GateMix, this steady source of income allows me to enjoy life to the fullest" (no period). There is a complete absence of strong personal voices, use of slang, inside jokes, sentence fragments, use of emojis, misspellings, off-topic discussions, disagreements, and spontaneous corrections that are all common in group chats. Terminology and punctuation is also oddly consistent. While some messages end in a question mark, not a single message ends with an exclamation mark or period. Several characters refer to "GateMix Manager" even though he never refers to himself by that term. Instead of using a $ to indicate U.S. dollars, every character uses USDT (e.g., "deposit 270 USDT", "30 USDT bonus"). It is difficult to definitively determine if a template is being used by humans creating these messages or if an LLM is being used. However, the consistency in tone, lack of misspellings, similarity of tone, consistent punctuation use, and variety of messages (none were duplicated in either group chat) suggest that an LLM is more likely.

H. Community Deception Techniques

Individual posts include a variety of common deception techniques such as appeal to authority (e.g., expert opinions of those who know about cryptocurrency investing), scarcity (e.g., urgent need to invest now), unity (e.g., feeling like part of the group and wanting to act like them), liking (e.g., a friendly character trying to help you), and especially social proof (e.g., hearing about how many others have successfully

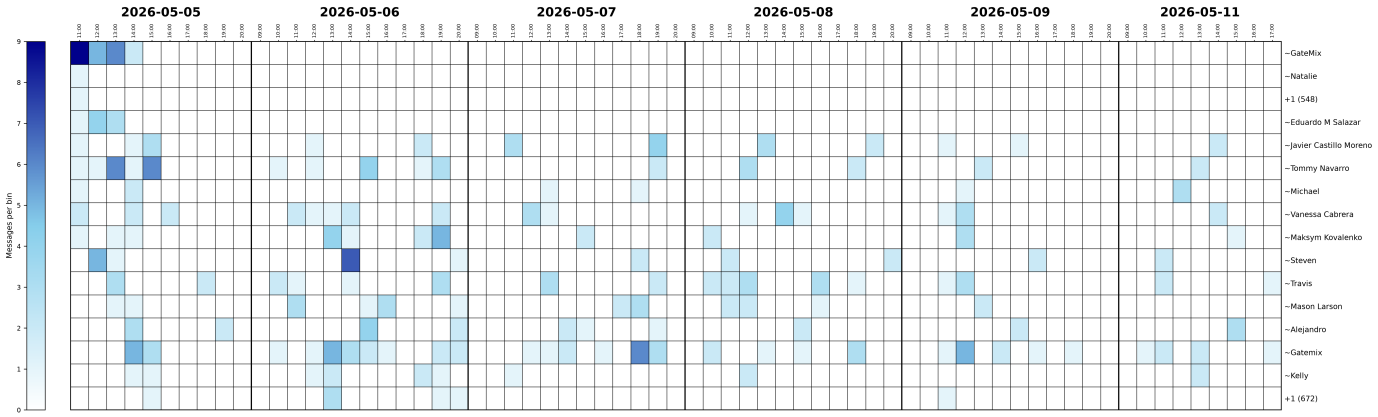


Fig. 6. Heatmap of GateMix Group 742 messages over time. No messages posted on Sunday, May 10.

made money). While these are well studied, the focus of this paper is on deception techniques that leverage a group of fake accounts. Table III summarizes these Community Deception Techniques, which rely upon a community of fake accounts. While they relate to specific social theories that are sometimes enacted by individuals (see Table III), they must be enacted by multiple participants in a group.

IV. Discussion

This study reviews a new scam approach that relies on coordinated communities of what appear to be fake accounts. While prior research has shown that scammers exploit principles such as authority, scarcity, liking, and especially social proof [5], [16], [19], our findings demonstrate how these principles become persuasive when enacted collectively. Rather than convincing victims through isolated messages, the GateMix WhatsApp groups create an entire social environment in which multiple fraudulent participants reinforce each other's credibility and normalize investment behavior.

These coordinated interactions resemble astroturfing and manufactured consensus [20]–[22]. However, instead of simply posting favorable testimonials, GateMix scammers fabricate a community whose members appear to ask questions, celebrate successes, mentor newcomers, and validate each other's experiences. Likewise, authority is distributed across complementary social roles rather than residing solely with the administrator, extending recent work on synthetic authority beyond individual personas to credibility that emerges from a coordinated network [23].

These observations motivate the concept of *community deception techniques*. Existing deception frameworks primarily describe persuasion strategies that can be executed by a single attacker [16], such as romance scams. In contrast, the techniques identified in this study depend on coordinated participation among multiple fraudulent accounts that collectively construct an artificial social reality. Dense network connections, frequent posting, and different social roles create the feeling of a thriving community. While this case study is limited to just two WhatsApp groups, the analysis provides a

TABLE III
Community Deception Techniques found in GateMix groups

Name	**Description and example**	**Related Theories**
Q&A Consensus Building	New members ask questions about cryptocurrency investing, and multiple experienced members respond with consistent advice recommending GateMix. Example: "Are you planning to invest in Bitcoin?... I recommend putting your funds into GateMix... I've already completed 11 orders..." followed by agreement from another member.	Consensus building; Social proof; Bandwagon effect; Appeal to popularity; Informational social influence
Community Exposure	Multiple members post success narratives in the form of screenshots showing profits and withdrawals or stories of successful investments over time. Example: "My order has been completed, and I'm planning to withdraw some of my profits..."	Mere-exposure effect; Social proof; Availability heuristic; Familiarity effects
Resolved Skepticism	One member expresses doubts or concerns and another community member resolves the concern. Example: "Crypto? I bought some before, but I lost money." Response: "Investing in crypto mixing is a 100% guaranteed way to make money..."	Uncertainty reduction; Informational social influence; Trust repair
Collective Validation of Authorities	Multiple members praise managers or experts, express gratitude, or attribute their success to guidance from authority figures. Examples: "Manager Jack, I'm delighted to have been invited..."; "This is all thanks to GateMix's compound interest investment..."	Authority principle; Social proof; Legitimation through consensus; Collective endorsement
Behavioral Modeling	Community members publicly describe taking increasingly committed actions (completing orders, investing more, inviting friends), encouraging others to imitate them. Example: newcomers who become experts like Tommy Navarro (discussed earlier); "I'm going to sign up for an account too."; "Gatemix's returns are so consistent. I definitely have to share this good news with my friends so they can join in too"	Social learning theory; Behavioral modeling; Descriptive norms; Injunctive norms; Social proof
Community Belonging	Members welcome newcomers, celebrate milestones, exchange greetings, and reinforce a shared group identity. Example: "Congratulations on joining GateMix!"; "Happy Saturday, everyone!"; "Hello everyone, I just finished making a deposit with James's help. I hope to learn and exchange knowledge about cryptocurrency trading with all of you"	Social identity theory; Sense of belonging; Group cohesion; In-group identification

template for analyzing other community level scams.

Finally, our findings suggest that community-level characteristics may provide valuable signals for scam detection. Complementary social roles, dense reciprocal interaction networks, rapid and consistent response times, and consistency in message formatting and tone across characters all may be used to identify scam communities. Using network metrics and temporal analyses may be particularly important as generative AI enables increasingly convincing text generation [18]. Future research should examine whether community deception techniques generalize across different messaging platforms, scam types, and AI-assisted operations, as well as investigate how authentic online communities differ structurally from fraudulent ones. Such work would help determine whether community-level behavioral indicators can support scalable detection and intervention strategies across emerging forms of coordinated social engineering.

V. Conclusion

This paper examined how coordinated WhatsApp cryptocurrency investment scams manufacture trust through the interaction of multiple fraudulent participants rather than a single scammer. Using a mixed-method case study approach employing qualitative analysis, temporal analysis, and social network analysis, we identified complementary social roles, coordinated interaction patterns, and network characteristics that create the appearance of a legitimate investment community. Building on these findings, we introduced the concept of *community deception techniques* to describe persuasion strategies that depend on coordinated participation among multiple fraudulent accounts. By drawing attention to community-level deception techniques, this work contributes to a broader understanding of coordinated social engineering and highlights opportunities for developing detection methods that leverage network structure, interaction patterns, and community behavior in addition to message content.

VI. Acknowledgments

This work was supported in part by NSF #2336409. ChatGPT-5.5 and Gemini 3.5 Flash were used for editing, formatting, and brainstorming titles.

References

[1] F. Salahdine and N. Kaabouch, "Social engineering attacks: A survey," *Future internet*, vol. 11, no. 4, p. 89, 2019.

[2] W. Syafitri, Z. Shukur, U. Asma'Mokhtar, R. Sulaiman, and M. A. Ibrahim, "Social engineering attacks prevention: A systematic literature review," *IEEE access*, vol. 10, pp. 39 325–39 343, 2022.

[3] Y. B. Choi and J. Rubin, "Social engineering cyber threats," *Journal of global awareness*, vol. 4, no. 2, p. 8, 2023.

[4] A. Ferreira and S. Teles, "Persuasion: How phishing emails can influence users and bypass security measures," *International Journal of Human-Computer Studies*, vol. 125, pp. 19–31, 2019.

[5] V. Luoma-aho, J. Botha, M. Hautala, S. J. Blackmon, and S. Karahan, "Vulnerabilities to crypto currency scams and online persuasion strategies," in *The Proceedings of the 19th International Conference on Cyber Warfare and Security*. Academic Conferences International Ltd, 2025.

[6] J. Botha, D. P. Botha, and L. Leenen, "An analysis of crypto scams during the covid-19 pandemic: 2020," in *ICCWS 2023 18th International Conference on Cyber Warfare and Security*. Academic Conferences and publishing limited, 2023.

[7] L. Pinjarkar, P. R. Hete, M. Mattada, S. Nejakar, P. Agrawal, and G. Kaur, "An examination of prevalent online scams: Phishing attacks, banking frauds, and e-commerce deceptions," in *2024 Second International Conference on Advances in Information Technology (ICAIT)*, vol. 1. IEEE, 2024, pp. 1–6.

[8] M. Apte, G. K. Palshikar, and S. Baskaran, "Frauds in online social networks: A review," *Social networks and surveillance for society*, pp. 1–18, 2018.

[9] M. A. Adebayo, "Role of social media in facilitating financial scams in abuja, nigeria: Development communication interventions for prevention," *African Journal of Stability and Development (AJSD)*, vol. 17, no. 1, pp. 631–648, 2025.

[10] Z. Guo, J.-H. Cho, R. Chen, S. Sengupta, M. Hong, and T. Mitra, "Online social deception and its countermeasures: A survey," *Ieee Access*, vol. 9, pp. 1770–1806, 2020.

[11] S. Mackenzie, "Criminology towards the metaverse: Cryptocurrency scams, grey economy and the technosocial," *The British Journal of Criminology*, vol. 62, no. 6, pp. 1537–1552, 2022.

[12] M. Dove, *The psychology of fraud, persuasion and scam techniques: Understanding what makes us vulnerable*. Routledge, 2025.

[13] B. Dulisse, C. Fitch, J. Denis, and N. Connealy, "Old techniques, new technologies: exploring patterns from scammers that commit financial fraud via cryptocurrency," *Journal of Economic Criminology*, vol. 9, p. 100178, 2025.

[14] M. McGuire, "Social media platforms and the cybercrime economy," *Bromium. Preuzeto*, vol. 21, p. 2023, 2019.

[15] M. Beluri, B. Acharya, S. Khodayari, G. Stivala, G. Pellegrino, and T. Holz, "Exploration of the dynamics of buy and sale of social media accounts," in *Proceedings of the 2025 ACM Internet Measurement Conference*, 2025, pp. 32–47.

[16] R. Cialdini and R. B. Cialdini, *Influence: Science and practice*. BoD–Books on Demand, 2013.

[17] J. McDowell, *Forensic Cyberpsychology and the Human Side of Cybercrime: The Mind Behind the Screen*. Routledge, 2026.

[18] M. T. Ashraf, "Synthetic trust attacks: Modeling how generative ai manipulates human decisions in social engineering fraud," *arXiv preprint arXiv:2604.04951*, 2026.

[19] N. L. Muscanell, R. E. Guadagno, and S. Murphy, "Weapons of influence misused: A social influence analysis of why people fall prey to internet scams," *Social and Personality Psychology Compass*, vol. 8, no. 7, pp. 388–396, 2014.

[20] J. Zhang, D. Carpenter, and M. Ko, "Online astroturfing: A theoretical perspective," in *Proceedings of the 2013 Americas Conference on Information Systems*, 2013.

[21] J. Chan, "Online astroturfing: A problem beyond disinformation," *Philosophy & Social Criticism*, vol. 50, no. 3, pp. 507–528, 2024.

[22] S. Woolley, *Manufacturing consensus: Understanding propaganda in the era of automation and anonymity*. Yale University Press, 2023.

[23] C. Rockwell and M. Conklin, "Synthetic authority: Ghost influencers, deepfakes, and the structural limits of disclosure-based financial regulation," *Deepfakes, and the Structural Limits of Disclosure-Based Financial Regulation (May 22, 2026)*, 2026.

[24] M. Muzammil, A. Pitumpe, X. Li, A. Rahmati, and N. Nikiforakis, "The poorest man in babylon: A longitudinal study of cryptocurrency investment scams," in *Proceedings of the ACM on Web Conference 2025*, 2025, pp. 1034–1045.

[25] M. T. Whitty and A. Sakzad, "The evolution of investment cyber scams: Vulnerability and victim blaming in the cryptocurrency era," *Computers & Security*, 2026.

[26] A. Fischer, "Manipulation and the affective realm of social media," in *The Philosophy of Online Manipulation*. Routledge, 2022, pp. 327–352.

[27] E. A. Gardner, G. Warner, S. Smith, N. Haines, W. Harris, and E. Adams, "Revealing the web of connections: Crypto scams, online drug sales, and financial crimes," *Forensic Science Review*, vol. 37, no. 2, pp. 88–95, 2025.

[28] R. K. Yin, *Case study methods*. American Psychological Association, 2012.

[29] D. Hansen, B. Shneiderman, and M. A. Smith, *Analyzing social media networks with NodeXL: Insights from a connected world*. Morgan Kaufmann, 2010.

[30] V. Clarke and V. Braun, "Thematic analysis," *The journal of positive psychology*, vol. 12, no. 3, pp. 297–298, 2017.